Powder Technology 203 (2010) 626–633

Contents lists available at ScienceDirect

Powder Technology

journal homepage: www.elsevier.com/locate/powtec

Radial porosity in packed beds of spheres

Gary E. Mueller *

Nuclear Engineering, Missouri University of Science and Technology, Rolla, MO 65409, USA

ARTICLE INFO

Article history:
Received 24 February 2010
Received in revised form 15 June 2010
Accepted 9 July 2010
Available online 17 July 2010

Keywords:
Porosity
Spheres
Packing
Cylinders
Radial
Analytical

ABSTRACT

A functional approach has been developed to investigate the radial porosity of mono-sized spheres in cylinders. Analytical and semi-analytical equations have been developed to calculate the local radial porosity and the radial porosity distribution, respectively, within a cylindrical packing structure. The analytical equations are based upon fundamental principles and are simple, straightforward and provide highly accurate results for the radial porosity with minimal computational prerequisites. The analytical equations have been developed for the fixed packing of identical spheres in cylindrical containers with $D/d \geq 2.0$. The predicted results for the local radial porosity and the radial porosity distributions are benchmarked with an existing analytical equation and available experimental data, respectively, for mono-sized spheres in cylindrical containers.

1. Introduction

The axially-averaged radial porosity variation is present in confined packed systems which are comprised of non-spherical or spherical particles. The radial porosity variation is a characteristic structural feature of confined fixed packed beds which occurs because of the influence of the container walls. It is universally recognized that the wall effects on the local porosity can be a significant constituent in the design and analysis of confined packed systems which include, for example, chromatography columns, chemical and nuclear reactors, and thermal heat exchangers. In particular, the radial porosity variations for mono-sized spheres in cylindrical containers are regularly used and have been investigated using various experimental and systematic methods by Roblee et al. [1]; Benenati and Brosilow [2]; Thadani and Peebles [3]; Ridgway and Tarbuck [4]; Martin [5]; Cohen and Metzner [6]; Goodling et al. [7]; Kufner and Hofmann [8]; Mueller [9]; Govindarao et al. [10]; Sederman et al. [11]; Wang et al. [12]; and Mariani et al. [13]. These experimental packing data provide an indispensable resource for benchmarking the accuracy of any type of predictive packing structural expression for spheres in cylindrical containers.

Packed bed hydrodynamic and heat transport numerical modeling techniques that analyze the bed as a pseudo-homogeneous media and therefore incorporate the local porosity variation generally involve using the axially-averaged radial porosity distribution. The local porosity variation in the radial direction, which is oscillatory and damped in nature, has been modeled completely by several different empirical approaches. Very simplified radial porosity models that characterize the radial porosity variation and use only an exponential-type function are typically similar to the models given by Vortmeyer and Schuster [14] and de Klerk [15]. The exponential-type radial porosity models do not account for the observed oscillations in the radial porosity and consequently are an average porosity as a function of the radial direction. The more representative radial porosity model developed by Martin [5]; Cohen and Metzner [6]; Kufner and Hofmann [8]; de Klerk [15]; and Bey and Eigenberger [16]; involves using a cosine function to characterize the radial oscillations and an exponential function to describe the radial damping. These model types predict the radial porosity very well near the cylindrical wall, but become less accurate as the radial distance from the wall increases. Mueller [9,17] characterized the oscillations with a Bessel function of the first kind of order zero and the radial damping with an exponential function. This model type over predicts the radial porosity near the cylindrical wall but describes the radial porosity accurately as the radial distance increases from the wall. In addition to the purely empirical models, more rigorous analytical approaches have been employed to characterize the radial porosity variation. Govindarao and Froment [18]; Govindarao and Ramrao [19]; and Govindarao et al. [20] developed expressions to describe the radial porosity variation with concentric cylindrical channels of equal thickness and spherical particle centers so that model accuracy increased with larger diameter aspect ratios. Du Toit [21] developed analytical based numerical procedures to evaluate area-based porosity. Mariani et al. [22–24] developed analytical closed expressions for the radial porosity profiles and the distribution of spherical particle centers. The comprehensive expressions of Mariani et al. [22]

* Tel.: +1 573 341 4348; fax: +1 573 341 6309.
E-mail address: gmueller@mst.edu.

0032-5910/$ – see front matter
doi:10.1016/j.powtec.2010.07.007

relate the intersection of a sphere and cylindrical surface in terms of natural functions that arise from integral calculus, i.e. elliptic integrals. The connection between the distribution of spherical particle centers and radial properties in random packed beds were also addressed by Mariani et al. [22], and consequently the results have significantly improved the understanding of this relationship. Nevertheless, even though the analytical expressions developed by Mariani et al. [22] are particularly accurate in predicting the radial porosity variation for spherical particles in cylindrical packed beds, the analytical predictive models have not been widely used in the engineering and scientific community in transport numerical modeling as compared to the less accurate and purely empirical radial porosity models.

As a result, there exists a significant disparity in the predictive radial porosity modeling domain between the less accurate purely empirical models and the more accurate complex analytical expressions. What is needed for transport numerical modeling for local phenomena that incorporates and accounts for the radial porosity variation is a model that is simple to use, accurate in the near and far wall regions, and analytical in nature, i.e., that it has been developed based on fundamental principles. It is for this reason, that an analytical based model can significantly improve the understanding of the packed bed structural configuration, which has been shown to be a function of the diameter aspect ratio, D/d, and decrease the errors propagated in the numerical simulations from using imprecise empirical radial porosity models.

The objective of this study is to present a functional approach based on established fundamental geometrical structural principles that has been developed to investigate the axially-averaged radial porosity variation of mono-sized spherical particles in cylindrical packed beds systems. Analytical expressions are presented for the local porosity variation and a semi-analytical expression for the radial porosity distribution. The new analytical expressions are compared with an existing analytical expression from the literature for the local radial porosity and validated with experimental data for the radial porosity distribution. The analytical expressions have been developed for fixed packed beds of identical spheres in cylindrical containers with $D/d \geq 2.0$.

2. Local radial porosity

Generally, the local radial porosity is characterized as a volumetric structural property of the packing system and has a numerical value between 0 and 1. Whether it is determined experimentally or through some analytical or numerical expression, it is represented as the axially-averaged local radial porosity that is obtained in a radial annular cylinder between the radii r and $r+\Delta r$. The local radial porosity, $\varepsilon(r)$, is normally defined as the fraction of the volume of voids to the total volume in the radial annular cylindrical layer. It is normally evaluated as $\varepsilon(r) = V_{void}/V_{total} = (V_{total} - V_{solid})/V_{total} = 1 - V_{solid}/V_{total}$, where V_{solid}, V_{void}, and V_{total} are the solid volume, void volume, and total volume, respectively, in the radial annular cylindrical layer. For a fixed cylindrical packed bed of mono-sized spherical particles of radius, R_s, with a center radial position of, r_s, this specific geometry condition in the x–y plane with an origin, O, is shown in Fig. 1. For this particular situation, the local radial porosity is expressed in terms of the solid volume contributions of spheres with radial centers positions within a particle radius on either side of the radial annular cylindrical layer. This established approach to calculating the local radial porosity has been used for well over 50 years. One of the principle priorities of this study is to first consider the radial annular cylindrical volume between the radii r and $r+\Delta r$ as shown in Fig. 1. For this local radial porosity analysis, let the radial annular cylindrical thickness, Δr, go to zero, $\Delta r = 0$, and as a result, what occurs is the intersection of a radial cylindrical surface of radius, r, from the origin, O, with that of any sphere with a radial center position, r_s, within a particle radius, R_s, on either side of this radial cylindrical surface. This particular geometrical situation is shown in Fig. 2. A somewhat comparable form of sphere–cylinder geometrical surface interaction has been considered and analyzed by Mariani et al. [22] together with geometrical expressions and quantities that were defined and derived in the investigation. For this current analysis, the local radial porosity, $\varepsilon(r)$, is now defined and evaluated in terms of this intersecting sphere–cylinder area as $\varepsilon(r) = A_{void}/A_{total} = (A_{total} - A_{solid})/A_{total} = 1 - A_{solid}/A_{total}$, where A_{solid}, A_{void}, and A_{total} are the solid area (intersecting area), void area (non-intersecting area), and total area (total radial cylindrical surface area), respectively, on the radial cylindrical surface. The local radial porosity, $\varepsilon(r)$, has now been reduced to obtaining expressions for areas instead of volumes and will result in a more uncomplicated and convenient expression for determining the local radial porosity. Hence, the local radial porosity, which is a function of the radial position, for a cylindrical packed bed system of mono-sized spheres is given by:

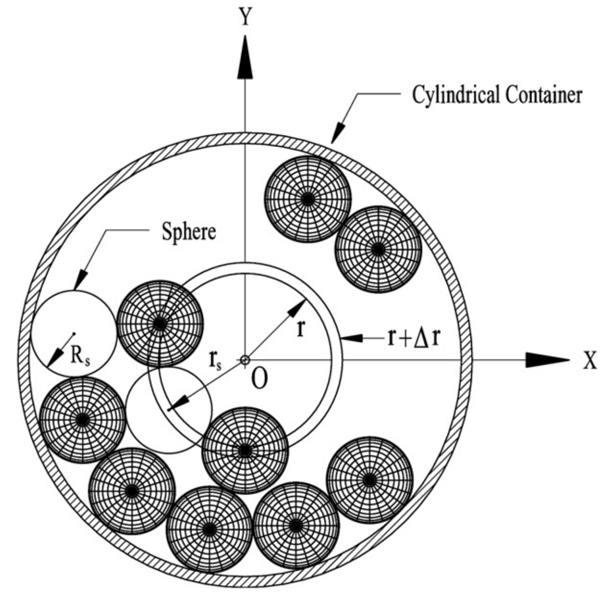

Fig. 1. X–Y plane of a radial annular layer at radial position r and $r+\Delta r$.

$$\varepsilon(r) = 1 - \frac{A_{solid}}{A_{total}} = 1 - \sum_{n=1}^{N} \frac{S_n(r)}{S_T(r)}, \tag{1}$$

where N is the number of sphere particles located within a sphere particle radius, R_s, on either side of the radial cylindrical surface at the radial position, r, $S_n(r)$ is the intersecting area of an nth-sphere at the radial location, r, and $S_T(r)$ is the total radial cylindrical surface area at a radial location, r, and is easily found to be $2\pi rH$ where H is the height of the radial cylindrical surface area.

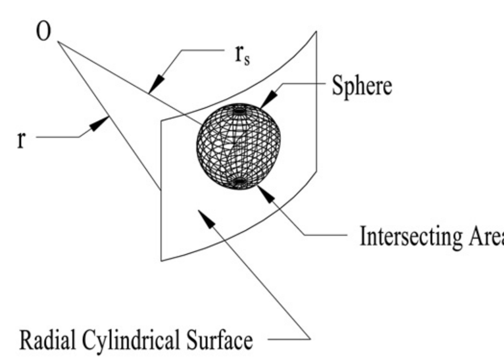

Fig. 2. Sphere intersecting area with radial cylindrical surface at radial position r.

The solid surface area, $S_n(r)$, is the area shown in Fig. 2 as the intersecting area of the radial cylindrical surface with that of a sphere. This intersecting solid area, $S_n(r)$, can be found easily by simultaneously solving the equations for y and z of a cylinder of radius, r, with its center at the origin, $x^2+y^2=r^2$, with that of a sphere of radius, R_s, with its center at a radial location along the x-axis of, r_s, $(x-r_s)^2+y^2+z^2=R_s^2$. The center radial location of any sphere in a cylindrical packing system can be transformed to have its center radial location along the x-axis by rotating the x–y axes about the origin and therefore this analysis can be used for all spheres in a packing system. These equations for a cylinder and sphere give the required two parametric equations for y and z for determining the intersecting area of the sphere and cylinder and are given as, respectively, $y=\pm[r^2-x^2]^{1/2}$ and $z=\pm[R_s^2-r^2+2xr_s-r_s^2]^{1/2}$. The intersecting surface area can be determined by calculating the arc length element around the intersecting surface area with the following equation, $ds=[1+(dy/dx)^2]^{1/2}\,dx$. The total intersecting sphere–cylinder surface area of an nth-sphere, $S_n(r)$, is a function of the radial cylindrical surface location, r, in a cylindrical packed bed system of radius, R_c, and can be determined by integrating the arc length with the appropriate upper limit, UL, and lower limit, LL, and the resulting equation is:

$$S_n(r) = \int_{LL}^{r} C\left[\frac{R_s^2 - r^2 + 2xr_s - r_s^2}{r^2-x^2}\right]^{1/2} r\,dx, \tag{2}$$

where

$$LL = \frac{r_s^2+r^2-R_s^2}{2r_s}, \quad \text{for } R_s\le r_s\le R_c-R_s, \text{ with } r_s-R_s\le r\le r_s+R_s, \text{ and } C=4,$$
$$\text{or } 0<r_s<R_s, \quad \text{with } R_s-r_s<r\le r_s+R_s, \text{ and } C=4, \tag{2a}$$

$$LL=-r, \quad \text{for } 0<r_s<R_s, \quad \text{with } 0\le r\le R_s-r_s, \quad \text{and } C=4, \tag{2b}$$

$$LL=0, \quad \text{for } r_s=0, \quad \text{with } 0\le r\le R_s, \quad \text{and } C=8. \tag{2c}$$

There are several different intersecting geometries that arise in a packed bed depending on the location of a sphere and the radial cylindrical surface. The different intersecting geometries will determine the specific expressions required for the lower limit of integration used in Eq. (2) which are given by Eqs. (2a)–(2c). The upper limit is the same for all cases and is given by, $UL=r$. The lower limit, LL, depends on the intersecting geometry type. The first intersecting geometry is for spheres with centers, r_s, that are at radial locations greater than or equal to a particle radius from the cylindrical container center, $R_s\le r_s\le R_c-R_s$ and the radial cylindrical surface with radial position, r, is $r_s-R_s\le r\le r_s+R_s$. Since the integration of the arc length is one quarter of the intersecting area, the constant, $C=4$, is required to obtain the total intersecting area. The majority of spheres in a cylindrical packed bed would be of this type of geometry and the lower limit is determined by Eq. (2a) and is given as, $LL=(r_s^2+r^2-R_s^2)/2r_s$. In addition, a similar geometry occurs when a sphere is at a radial location less than a particle radius from the center of the container, $0<r_s<R_s$ with $R_s-r_s<r\le r_s+R_s$, and once again the lower limit is determined from Eq. (2a) and is given as, $LL=(r_s^2+r^2-R_s^2)/2r_s$. To obtain a solution for these first geometry types, Eq. (2) must be evaluated numerically with the stated lower and upper limits. The second type of geometry takes place when the center of a sphere is at a radial location less than a particle radius from the center of the cylindrical container, $0<r_s<R_s$, and the radial cylindrical surface with radial position is such that, $0\le r\le R_s-r_s$. The lower limit for this type of geometry is determined from Eq. (2b) and is given as, $LL=-r$. For the second geometry type, the integration of the arc length is one quarter of the intersecting area and the constant, $C=4$, is required to obtain the total intersecting area. Again, a solution for the second type of geometry position is determined numerically from Eq. (2) with the stated lower and upper limits. The final type of intersecting geometry occurs when the radial center position of the sphere is at the center of the cylindrical container, $r_s=0$, and the radial cylindrical surface with radial position is, $0\le r\le R_s$. The lower limit of integration of Eq. (2) is determined from Eq. (2c) and is given as, $LL=0$. For this geometry type, the integration of the arc length is one eighth of the intersecting area and the constant, $C=8$, is required to obtain the total intersecting area. An analytical closed solution is obtained for this geometry type from Eq. (2) with the limit of Eq. (2c) and the total intersecting area is $S_n(r)=4\pi r(R_s^2-r^2)^{1/2}$.

Eq. (2) is an exact analytical expression for determining the intersecting area of an nth-sphere in a cylindrical packed bed system, and when combined with Eq. (1), can be used to analytically determine the local radial porosity, $\varepsilon(r)$, as a function of the radial position, r. The only limitations imposed for the use of Eqs. (1) and (2) for determining the local radial porosity are that the spheres must be mono-sized in shape and packed in a cylindrical bed. Even though Eq. (2) is an exact expression and practical for determining the intersecting area, it still involves an integral which leads to the question as to if there is an approximation to Eq. (2) which can yield a sufficient accurate solution while eliminating the need to evaluate the integral expression. To answer the question of determining an approximate expression for Eq. (2), Fig. 2 should be considered again. As the radial position, r, of the radial cylindrical surface increases, the curvature, κ, of the surface decreases since it is inversely proportional to the radial position. In the limit as $r\to\infty$, $\kappa\to 0$, and the radial cylindrical surface becomes a plane at the radial position, r. As a result, the solid surface area, $S_n(r)$, in Fig. 2 is now the intersecting area of a plane with that of a sphere in the y–z plane. This intersecting solid area, $S_n(r)$, can now be determined easily by simultaneously solving the equations for a plane at the position, r, with its center at the origin, $x=r$, with that of a sphere of radius, R_s, with its center at a radial location along the x-axis, r_s, $(x-r_s)^2+y^2+z^2=R_s^2$. Substituting the equation for the intersecting plane, $x=r$, into the equation of the sphere results in the following equation, $y^2+z^2=R_s^2-(r-r_s)^2$. This is the equation of a circle in the y–z plane with a radius of $[R_s^2-(r-r_s)^2]^{1/2}$ and an intersecting area of $S_n(r)=\pi[R_s^2-(r-r_s)^2]$ and is listed below as Eq. (3a) and is the approximate expression of Eq. (2a) with the same domain. The majority of spheres in a cylindrical packed bed would be of this type of geometry, as previously indicated. The next approximate expression is for Eq. (2b), when the center of a sphere is at a radial location less than a particle radius from the center of the cylindrical container, $0<r_s<R_s$, and the radial cylindrical surface with radial position is such that, $0\le r\le R_s-r_s$. The approximate expression for Eq. (2b) is easily determined from the surface area of a cylinder with its center and its boundaries within that of a sphere and is below listed as Eq. (3b) with the same domain as Eq. (2b). When $r\le r_s$ the cylinder surface area that is truncated by a sphere surface is $S_n(r)=2\pi r\{[R_s^2-(r+r_s)^2]^{1/2}+[R_s^2-(r-r_s)^2]^{1/2}\}$ and is listed below as the first expression for Eq. (3b). When $r>r_s$ the cylinder surface area truncated by a sphere surface is given by $S_n(r)=\pi r\{[R_s^2-(r+r_s)^2]^{1/2}+3(R_s^2-r_s^2)^{1/2}\}$, which tends to overestimate the surface area. The first expression for Eq. (3b) can be used for $r>r_s$ and it tends to underestimate the surface area. Therefore, averaging the two expressions provides a good result for the surface area for $r>r_s$ and is listed below as the second expression for Eq. (3b). The final equation is an analytical closed expression and therefore no approximate expression is required since the radial center position of the sphere is at the center of the cylindrical container and is given by Eq. (2c).

$$S_n(r) = \pi[R_s^2-(r-r_s)^2], \text{ for } R_s\le r_s\le R_c-R_s, \text{ with } r_s-R_s\le r\le r_s+R_s,$$
$$\text{or } 0<r_s<R_s, \quad \text{with } R_s-r_s<r\le r_s+R_s, \tag{3a}$$

$$S_n(r) = 2\pi r\left\{\left[R_s^2-(r+r_s)^2\right]^{1/2}+\left[R_s^2-(r-r_s)^2\right]^{1/2}\right\}, \text{when } r \le r_s,$$
$$S_n(r) = \tfrac{1}{2}\pi r\left\{3\left[R_s^2-(r+r_s)^2\right]^{1/2}+2\left[R_s^2-(r-r_s)^2\right]^{1/2}+3\left(R_s^2-r_s^2\right)^{1/2}\right\},$$
$$\text{when } r > r_s,\quad \text{for } 0 < r_s < R_s,\quad \text{with } 0 \le r \le R_s - r_s. \tag{3b}$$

3. Radial porosity distribution

The radial porosity distribution or radial porosity profile is the variation of the local radial porosity as a function of the radial direction and can be determined from Eq. (1). To generate the radial porosity profile a distribution of the sphere particle centers in the cylindrical packed bed is required. The distribution of the sphere particle centers can be specified from experimentally determined values or from a distribution function expression. To evaluate Eq. (1), the experimental measured data from Mueller [9] is used for the packed bed with the diameter aspect ratio of $D/d = 7.99$. Fig. 3 shows the results for this investigation using the analytical expressions of Eqs. (1) and (2) and the analytical expression of Eq. (1) and the approximate analytical expression of Eq. (3), Eqs. (3a) and (3b) combined. In addition, a comparison is made between the area-derived radial porosity profile and Mueller's [9] volume-derived radial porosity profile. As seen in Fig. 3, the area-derived radial porosity profile of the analytical expressions of Eqs. (1) and (2) is identical with Mueller's [9] volume-derived radial porosity profile. The important advancement is that the area-derived radial porosity profile analytical expressions of Eqs. (1) and (2) are far less complicated and considerably easier to apply than volume-derived radial porosity profile expressions, which can be rather complex and cumbersome to apply. What is surprising is the result of the area-derived radial porosity profile from the analytical expression of Eq. (1) and the approximate analytical expression of Eq. (3). It has been established that replacing the radius r with an effective radius, $r_e = (UL + LL)/2 = [(r_s + r)^2 - R_s^2]/4r_s$, where $UL = r$ and $LL = (r_s^2 + r^2 - R_s^2)/2r_s$ are the limits of Eq. (2a), provides a more accurate fit for Eq. (3a). As seen in Fig. 3, the approximate analytical expression profile is almost identical to the two exact analytical expression profiles up to about one-half of a diameter from the center of the packed bed. This is quite significant since most investigators are generally more interested in the radial regions next to the cylindrical wall and not in the center region of the packed bed system. There is some small variance in the approximate profile in the last one-half of a diameter from the center of the packed bed. This is due to the fact that the curvature is important at these small radial positions and that it affects the results obtained from the approximate radial porosity expressions in Eq. (3). In addition, Table 1 (Estimate of standard error of regression) illustrates the precision of prediction in percent of the individual approximate Eqs. (3a) and (3b) and the combined two equations as Eq. (3) compared with Eqs. (2a) and (2b) and the combined two equations as Eq. (2), respectively, for ten different D/d ratios ranging from 2.0 to 20.0 using the sphere center data of Govindarao et al. [10] and Mueller [9,25]. The estimate of standard error of regression illustrates how well the regression curve approximates the real data and is a common goodness of fit measurement used in curve-fitting statistics. As shown in Table 1, the approximate Eq. (3a) is a very good substitute for the analytical expression of Eq. (2a) of the radial profile for D/d ratios greater than or equal to 3.00. There are two values listed for Eq. (3a) for each D/d ratio. The first listed value is for the complete domain of Eq. (3a),$R_s \le r_s \le R_c - R_s$, with $r_s - R_s \le r \le r_s + R_s$, and $0 < r_s < R_s$ with $R_s - r_s < r \le r_s + R_s$. This includes the wall region and the region near the center of the packed bed when $0 < r_s < R_s$. The second listed value in parentheses is for only the first portion of the domain of Eq. (3a), $R_s \le r_s \le R_c - R_s$, with $r_s - R_s \le r \le r_s + R_s$. This includes the wall region but does not include the region near the center of the packed bed when $0 < r_s < R_s$. As shown by these two values, the majority of the error for the approximate Eq. (3a) is accumulated in the region near the center of the packed bed when $0 < r_s < R_s$. This is because in this region the curvature is important and the plane approximation used in Eq. (3a) is insufficient to accurately represent the curvature. As shown in Table 1 the approximate Eq. (3b) is an excellent substitute for the analytical expression of Eq. (2b) for the domain $0 < r_s < R_s$, with $0 \le r \le R_s - r_s$ in the region near the center radial profile for all D/d ratios. In addition, there are two D/d ratios listed Table 1 that require additional clarification. The errors for Eqs. (3a) and (3b) for the D/d ratios of 3.00 and 7.99 are relatively large as compared with their combined values listed for Eq. (3). For example, the 7.99 D/d ratio has errors of 1.17 (0.56), 2.48 and 0.76 for Eqs. (3a), (3b) and (3), respectively. The reason why the combined Eq. (3) has a much smaller error value (0.76) is that Eq. (3a) over predicts the radial porosity in the domain $0 < r_s < R_s$ as given by its errors of 1.17 (0.56) and that Eq. (3b) under predicts the radial porosity in the domain $0 < r_s < R_s$ as given by its error of 2.48 and the combination of the two equations results in a very small error. As a result, Eqs. (3), (3a) and (3b) combined, as shown in Table 1 provides very good results for D/d ratios greater than or equal to 3.00 and is a very good approximate equation for Eq. (2).

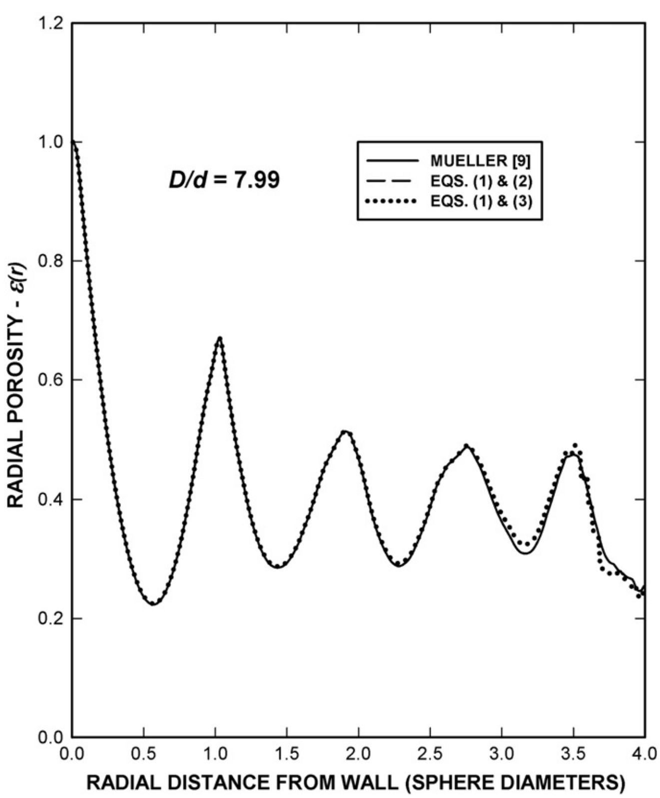

Fig. 3. Radial porosity comparison between analytical expressions.

As shown, radial porosity distributions or radial porosity profiles can be generated using the local radial porosity given the sphere particle centers. As a result, the analytical expressions of Eqs. (1) and (2) or Eqs. (1) and (3) can be used for this purpose for any cylindrical packed bed with mono-sized spheres. It has been shown that Eqs. (1) and (3a) provide almost identical results for the majority of the radial porosity profile and since it is a simple equation it will be used to develop a semi-analytical expression for modeling the radial porosity profiles. The challenge in modeling the radial porosity profiles with Eqs. (1) and (3a) is obtaining an appropriate function which characterizes the distribution of the mono-sized spherical particle centers in a packed bed cylindrical system. To gain an insight into the nature of this function it is necessary to investigate how the sphere particle centers are distributed in a cylindrical packed bed. A typical example of this sphere particle center distribution in the x–y plane is shown in Fig. 4 and is from the experimental data obtained by Mueller [9] for a cylindrical packed bed with a diameter aspect ratio of $D/d = 7.99$. Mariani et al. [13] presents a comparable example of the distribution of sphere particle centers in a cylindrical packed bed for a diameter aspect ratio of $D/d = 5.04$. In each case, the first layer section adjacent to the wall has a high concentration of spheres and is also highly ordered with almost all of the sphere particle centers having a value of one-half of a particle diameter. The second layer section continues to have a high concentration of spheres but shows a definite disorder in the distribution of the radial sphere particle centers. This development of a decreasing concentration of spheres and decreasing order in the sphere particle centers from the cylindrical wall continues toward the center of the cylindrical packing system. Mariani et al. [24] describes that the distribution of particles centers can be described by a series of zones or layer sections with either concentrated particle centers or uniformly distributed particle centers. They also indicated that sphere particle centers typically have been defined mathematically in a particular zone with either an impulse function for particle centers ideally concentrated at one position or with a step function for particle centers that are uniformly distributed. Mariani et al. [24] and previous investigations in the open literature conclude that these types of distributions are suitable for all practical purposes and even though there is extensive information about randomly packed, uniformly sized spheres, predictive expressions can still be improved upon.

The particular function that will be used in this investigation to distinctively account for the distribution of sphere particle centers in the cylindrical packed system is the floor function, $\lfloor x \rfloor$, which is also known as the greatest integer function or integer value function and it provides the largest integer less than or equal to x and is a specific type of step function [26]. An equivalent representation of the floor function using elementary functions is given by, $\lfloor x \rfloor = x + (\tan^{-1}[\cot(\pi x)])/\pi - 1/2$. Mariani et al. [24] have established that a step function is a suitable function for representing the distribution of sphere particle centers in packed cylindrical systems. As shown in Fig. 4, almost all the sphere particle centers in the first layer section adjacent to the wall are positioned at one-half of a sphere particle diameter. No sphere particle centers can exist in the region between the wall and one-half of a sphere particle diameter for mono-sized spherical particles. For an ideal packed bed, no sphere particle centers would exist between the first and second layer sections and continue throughout the cylindrical packing system. Even in an actual packed bed as shown in Fig. 4, only a small number of sphere particle centers exist in the regions between the major particle center ring positions. This typical packing distribution of the sphere particle centers where the majority of the centers concentrate at particular radial positions indicates a discontinuity in the distribution of the sphere particle centers. This is the principal motivation for using the floor function to model the distribution of the sphere particle centers.

To establish the use of the floor function in characterizing the sphere particle center distribution for a radial porosity profile correlation the experimental fixed packed bed sphere data from Mueller [9] for a diameter aspect ratio, $D/d = 7.99$, is used as a comparison to illustrate the procedure for developing the radial porosity semi-analytical

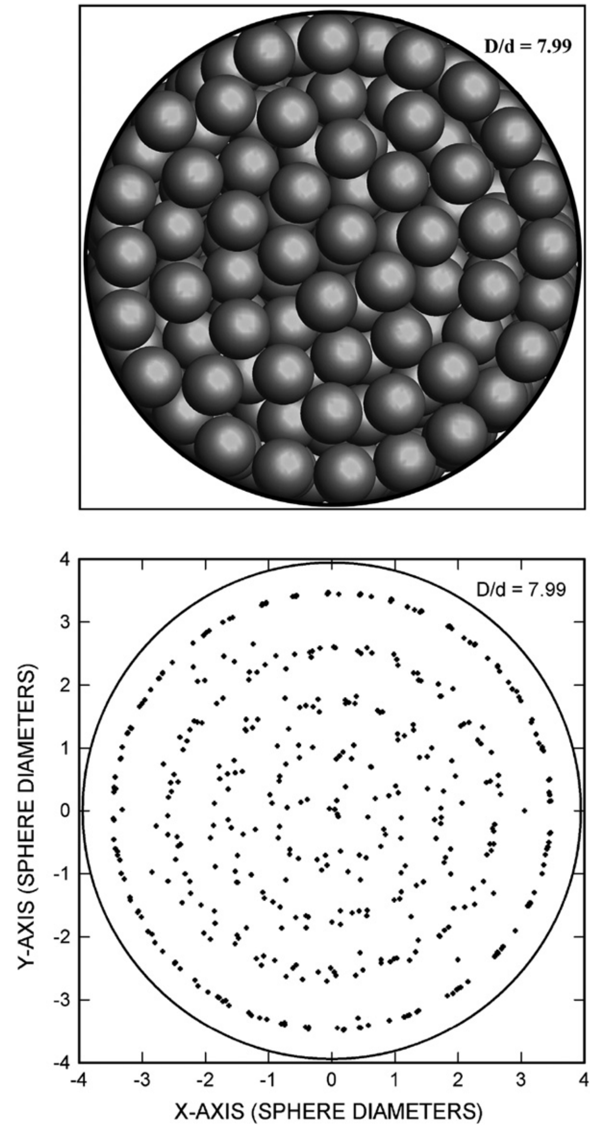

Fig. 4. Sphere particle center radial distribution for $D/d = 7.99$, Mueller [9].

expression. The analytical expressions of Eqs. (1) and (3a) are used to formulate the general composition of the principal equation for the radial porosity semi-analytical expression which is given as $\varepsilon(r') = 1 - \sum S_n(r')/S_T(r') = 1 - \sum \pi[R_s^2 - (r' - r_s)^2]/S_T(r') = 1 - \pi[R_{se}^2 - \alpha(r_e - r_{se})^2]$, where r' is the sphere diameter normalized radial position from the wall, $R_{se}^2 = \sum R_s^2/S_T(r')$ and $\alpha(r_e - r_{se})^2 = \sum (r' - r_s)^2/S_T(r')$. The general expression of the principal equation now includes the new parameters R_{se}, α, r_e, and r_{se} which have to be analytically or empirically determined. Initially assuming $R_{se} = R_s^* = 1/2$, where R_s^* is the normalized sphere radius, $\alpha = 1$, and using an ideal sphere packing as previously described where the particle centers are specifically distributed at $1R_s^*$, $3R_s^*$, etc., the floor function can be used to analytically characterize the sphere radial particle center position distribution with

Table 1
Estimate of standard error of regression in percent.

	Diameter aspect ratio, D/d									
Equation	2.00	3.00	3.96	5.00	5.96	7.00	7.99	10.0	14.1	20.3
(3a)	8.67 (3.13)	2.06 (0.87)	2.97 (2.24)	3.02 (0.53)	1.83 (0.95)	2.33 (0.68)	1.17 (0.56)	1.84 (0.44)	1.35 (0.38)	1.16 (0.34)
(3b)	0.59	2.91	0.14	0.73	0.47	0.59	2.48	0.65	0.80	0.62
(3)	8.62	0.92	2.95	2.91	1.79	2.31	0.76	1.77	1.19	1.08

$r_{se} = \lfloor r_r + R_s^* \rfloor$ where the radial position parameter, $r_r = r' + R_s^*$ translates the radial porosity profile by a sphere radius, R_s^*, so that the maximum occurs at the origin. The profile can be easily determined for the radial porosity, $\varepsilon(r') = 1 - \pi[R_{se}^2 - \alpha(r_e - r_{se})^2]$, with the effective radius $r_e = [(r_{se} + r_r)^2 - R_s^{*2}]/4r_{se}$ and is shown in Fig. 5(a). The dotted curve is the Mueller [9] data radial porosity profile and the line curve is the ideal radial porosity profile, $\varepsilon(r')$ for $D/d = 7.99$. To damp the oscillations in the radial porosity profile and account for the distribution in the particle center locations about the ideal center positions of $1R_s^*$, $3R_s^*$, etc. the parameter α is empirically adjusted from $\alpha = 1$ to $\alpha = 1/(cr'^{\,1.4} + 1)$, where $c = 0.745$ and the results are shown in Fig. 5(b). The horizontal location of the minima and maxima in the profile are empirically adjusted by changing $r_r = r' + R_s^*$ to $r_r = br'^2 + r' + R_s^*$, where $b = 0.036$ with the result shown in Fig. 5(c). The final empirical adjustment for the semi-analytical expression changes $R_{se} = R_s^*$ to $R_{se} = R_s^* - ar'^2/(r'^2 + 2)$, where $a = 0.045$ and the results are shown in Fig. 5(d). This case for $D/d = 7.99$ establishes the final form of the semi-analytical expression for the radial porosity correlation. The empirical parameters for a, b, and c are determined from experimental data and new empirical equations are developed for each parameter.

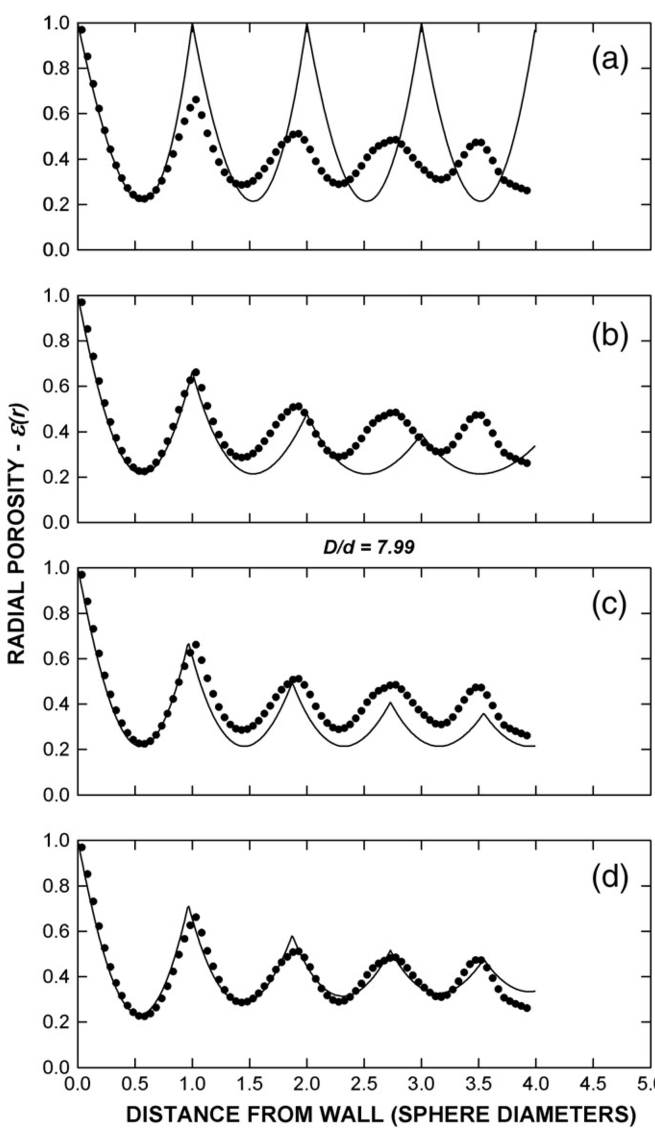

Fig. 5. Radial porosity profile model development.

The radial porosity semi-analytical expression at a sphere diameter normalized radial position from the wall, r', for mono-sized unit spheres in a cylindrical packed system is:

$$\varepsilon(r') = 1 - S_e(r'), \tag{4}$$

where

$$S_e(r') = \pi[R_{se}^2 - \alpha(r_e - r_{se})^2], \tag{4a}$$

and

$$R_{se} = R_s^* - \frac{ar'^2}{r'^2 + 2}, \text{ with } a = 0.045 - 0.55e^{-D/d}, \tag{4b}$$

$$r_e = \frac{(r_{se} + r_r)^2 - R_s^{*2}}{4r_{se}}, \tag{4c}$$

$$r_{se} = \lfloor r_r + R_s^* \rfloor, \tag{4d}$$

$$r_r = br'^2 + r' + R_s^*, \text{ with } b = 0.037 - 2.0e^{-D/d}, \tag{4e}$$

$$\alpha = \frac{1}{cr'^{1.4} + 1}, \text{ with } c = 1.83 - \frac{19.2}{D/d} + \frac{40.5\ln(D/d)}{(D/d)^2}, \tag{4f}$$

with $R_s^* = ½$, $d^* = 1.0$, $0 \le r' \le R_c/d$, $R_c = D/2$, and $D/d \ge 2.0$.

4. Results and discussion

The expanded presentation of the semi-analytical expression, Eq. (4) for the radial porosity distribution is implemented to elucidate the specific terms involved in the equation. Eq. (4) is valid for mono-sized unit spheres in fixed, packed cylindrical beds with diameter aspect ratios of $D/d \ge 2.0$. The mono-sized spheres are normalized unit spheres with diameters and radii equal to, $d^* = 1.0$, $R_s^* = 1/2$, respectively. This permits Eq. (4) to be used for any mono-sized sphere type, cylindrical packing system since the sphere diameters and radii can be easily normalized to, $d^* = d/d = 1.0$ and $R_s^* = R_s/d = 1/2$, respectively, with a cylindrical packing diameter equal to D/d. The significance of the semi-analytical expression of Eq. (4) is that it has the correct mathematical form for the radial porosity since it was derived from the analytical expressions of Eqs. (1) and (3a). The effective intersecting area, $S_e(r')$, of Eq. (4), and specified by Eq. (4a), corresponds to the summation term of Eq. (1). This effective intersecting area represents the ratio of the entire intersecting solid area with the total area at a normalized radial position, r'. The effective intersecting area, $S_e(r')$, is given by Eq. (4a) and has the same mathematical form as the approximate analytical expression of Eq. (3a). The first term in the expression is the effective sphere radius, R_{se}, which is given by Eq. (4b) and corresponds to the sphere radius of Eq. (3a). It has been shown that the concentration of spheres decreases from the wall to the center of the cylindrical packed bed. The effective sphere radius represents all the spheres at a particular radial position as an effective sphere and compensates for its radius as a function of the radial position. The next expression in Eq. (4a) is the effective radial position, r_e, given by Eq. (4c) which corresponds to the radial position of Eq. (3a). As previously stated the effective radius, $r_e = (UL + LL)/2 = [(r_s + r)^2 - R_s^2]/4r_s$, where $UL = r$ and $LL = (r_s^2 + r^2 - R_s^2)/2r_s$, and for Eq. (4c) r_s has been replaced with r_{se}, r with r_r, and R_s with R_s^*. The last term in Eq. (4a) is the effective sphere center radial position, r_{se}, and corresponds to the radial position of the sphere centers, r_s, in Eq. (3a). The effective sphere center radial position, r_{se}, is determined from Eq. (4d) and involves the floor function, which is used to assist in modeling the distribution of sphere centers that occur at particular radial positions as shown in Fig. 4. The radial position parameter, r_r, given by Eq. (4e) is also incorporated to help account for the sphere center positions. An empirical parameter in Eq. (4a), α, given by Eq. (4f) is used to adjust the damping of the observed oscillations of the radial porosity distribution. Three empirical parameters, a, b, and c, are used to take into account the variations in the radial porosity distribution as a function of the cylindrical packed bed diameter aspect ratio, D/d. The parameter a regulates the vertical position of the oscillations; b adjusts the frequency of oscillations; and c affects the amplitude of the oscillations. The semi-analytical expression for the radial porosity profile, Eq. (4) has been compared with existing experimental data and the results are shown in Fig. 6. As shown, the semi-analytical model correctly predicts the radial porosity in the region at the wall and in the region far from the wall for small and large diameter aspect ratios.

In conclusion, analytical equations, Eqs. (1)–(3), have been presented to determine the local radial porosity for mono-sized spheres in fixed cylindrical packed systems. From these analytical expressions, a semi-analytical model, Eq. (4) has been developed for the radial porosity that can be used for a cylindrical packed system with diameter aspect ratios of $D/d \geq 2.0$. The semi-analytical equation accurately predicts the radial porosity in the near and far wall regions and has the correct mathematical form for determining the radial porosity.

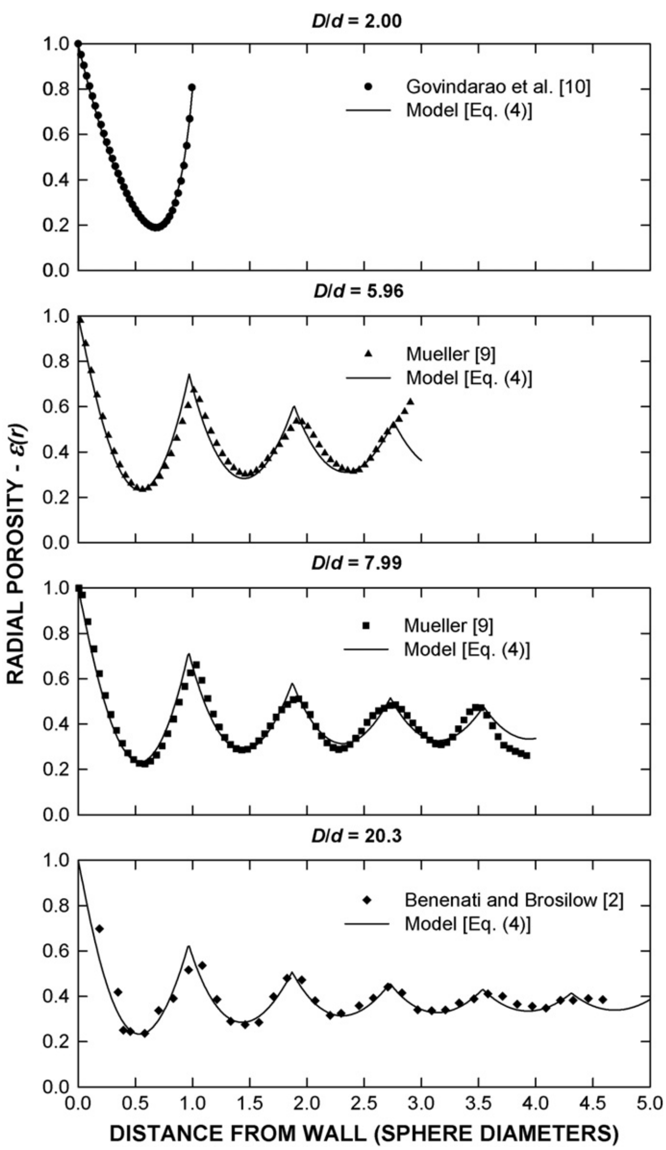

Fig. 6. Radial porosity comparison between analytical model and experimental data.

List of symbols

a	parameter, Eq. (4b)
A_{solid}	intersecting area (m^2)
A_{total}	total area (m^2)
A_{void}	void area (m^2)
b	parameter, Eq. (4e)
c	parameter, Eq. (4f)
C	constant
d	sphere diameter (m)
ds	arc length (m)
dx	differential length (m)
dy	differential length (m)
d^*	normalized sphere diameter, d/d
D	cylinder diameter (m)
H	cylinder height (m)
LL	lower limit
n	index variable
N	number of sphere particles
O	origin
r	radial variable (m)
r_e	effective radius
r_r	radial position parameter
r_s	sphere center radial position (m)
r_{se}	effective sphere center radial position
r'	normalized radial position, $(R_c - r)/d$
Δr	radial annular layer thickness (m)
R_c	cylinder radius (m)
R_s	sphere radius (m)
R_{se}	effective sphere radius
R_s^*	normalized sphere radius, R_s/d
S_e	effective intersecting area
S_n	intersecting sphere–cylinder area (m^2)
S_T	total radial cylindrical surface area (m^2)
UL	upper limit
V_{solid}	solid volume (m^3)
V_{total}	total volume (m^3)
V_{void}	void volume (m^3)
x	coordinate (m)
y	coordinate (m)
z	coordinate (m)

Greek letters

α	parameter, Eq. (4f)
ε	radial porosity
κ	curvature
π	pi

References

[1] L.H.S. Roblee, R.M. Baird, J.W. Tierney, AIChE J. 4 (1958) 460.
[2] R.F. Benenati, C.B. Brosilow, AIChE J. 8 (1962) 359.
[3] M. Thadani, F.N. Peebles, Ind. Eng. Chem. Proc. Des. Dev. 5 (1966) 265.
[4] K. Ridgway, K.J. Tarbuck, Chem. Eng. Sci. 23 (1968) 1147.
[5] H. Martin, Chem. Eng. Sci. 33 (1978) 913.
[6] Y. Cohen, A.B. Metzner, AIChE J. 27 (1981) 705.
[7] J.S. Goodling, R.I. Vachon, W.S. Stelpfug, S.J. Ying, M.S. Khader, Power Technol. 35 (1983) 23.
[8] R. Kufner, H. Hofmann, Chem. Eng. Sci. 45 (1990) 2141.
[9] G.E. Mueller, Powder Technol. 72 (1992) 269.
[10] V.M.H. Govindarao, K.V.S. Ramrao, A.V.S. Rao, Chem. Eng. Sci. 47 (1992) 2105.
[11] A.J. Sederman, P. Alexander, L.F. Gladden, Powder Technol. 117 (2001) 255.
[12] Z. Wang, A. Afacan, K. Nandakumar, K.T. Chuang, Chem. Eng. Proc. 40 (2001) 209.

[13] N. Mariani, W. Salvat, A. Campesi, G. Barreto, O. Martínez, Int. J. Chem. React. Eng. 7 (2009) 1.
[14] D. Vortmeyer, J. Schuster, Chem. Eng. Sci. 38 (1983) 1691.
[15] A. de Klerk, AIChE J. 49 (2003) 2022.
[16] O. Bey, G. Eigenberger, Chem. Eng. Sci. 50 (1997) 1365.
[17] G.E. Mueller, Chem. Eng. Sci. 46 (1991) 706.
[18] V.M.H. Govindarao, G.F. Froment, Chem. Eng. Sci. 41 (1986) 533.
[19] V.M.H. Govindarao, K.V.S. Ramrao, Chem. Eng. Sci. 43 (1988) 2544.
[20] V.M.H. Govindarao, M. Subbanna, A.V.S. Rao, K.V.S. Ramrao, Chem. Eng. Sci. 45 (1990) 362.
[21] C.G. Du Toit, Proc. 1st Int. Topical Meeting HTRT, Petten, Netherlands, 2002.
[22] N.J. Mariani, G.D. Mazza, O.M. Martinez, G.F. Barreto, Trends Heat Mass Mom. Tran. 4 (1998) 95.
[23] N.J. Mariani, G.D. Mazza, O.M. Martinez, G.F. Barreto, Can. J. Chem. Eng. 78 (2000) 1133.
[24] N.J. Mariani, O.M. Martinez, G.F. Barreto, Chem. Eng. Sci. 56 (2001) 5693.
[25] G.E. Mueller, Powder Technol. 159 (2005) 105.
[26] E.W. Weisstein, MathWorld (2010), http://mathworld.wolfram.com/FloorFunction.html.